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Self-Playing Orchestra

Milestones

1. MIDI Communication / UART Interface

Objective:

Develop and verify communication between the PC and microcontroller. The system will receive MIDI data from the PC through UART and correctly read the incoming data.

Verification:

This test shows MIDI commands sent from the PC to the MCU. The status byte, data byte 1, and data byte 2 will be displayed on a serial terminal.

Measurable Outcome:

Send a known sequence of MIDI commands and verify that 100% of the transmitted commands are received and decoded correctly.

2. Twelve Solenoid MIDI Showcase

Objective:

We will interface all twelve solenoids with the microcontroller and activate the correct solenoid based on the received UART MIDI messages.

Verification:

Each of the twelve MIDI notes will be closely observed and subsequently verified that each corresponding solenoid activates and releases. We will use a MIDI piano to test each note individually.

Measurable Outcome:

All twelve solenoids must respond to their assigned MIDI notes accurately, with no incorrect solenoid activations during testing.

3. Wind Box & Sound Production

Objective:

Design and build a wind box for the pipe organ that provides controlled airflow to said instrument. The system will use the wind source together with the solenoid-controlled notes so that activating a solenoid produces the correct audible musical note.

Verification:

We will test each of the twelve solenoids by activating them individually while the wind box is running. We need to verify that airflow reaches the pipe organ and that each activated solenoid produces a clear and recognizable sound.

Measurable Outcome:

We successfully produce sound from all 12 notes individually, with each solenoid generating its intended note when activated. We verify that all 12 notes can be played clearly and that the wind box can maintain sufficient airflow while playing.

4. MIDI Note to Instrument Integration

Objective:

Integrate MIDI decoding, solenoid control, and the wind system, so MIDI notes received from the computer play the corresponding notes.

Verification:

This test will send individual MIDI notes from the pc, and the mcu will decode each note and activate the correct solenoid which will physically produce the corresponding sound.

Measurable Outcome:

Test different supported MIDI notes. The correct mechanical note must be activated for and successfully produce sound.

5. Play MIDI Songs

Objective:

Integrate the MIDI communication, solenoid control, and wind box into one complete system. The system will receive a MIDI song from the PC and automatically play the corresponding notes on the instrument.

Verification:

We send a selected MIDI song from the PC to the microcontroller. We then observe that the correct solenoids activate at the correct times and that the wind box provides enough airflow for the notes to be heard throughout the song.

Measurable Outcome:

We successfully play a complete MIDI song from beginning to end. We verify that the system responds to the MIDI commands, plays the correct notes in sequence, and completes the song without missed commands or system failures.